

BUILD YOUR PERSONAL CYBERSECURITY LAB

Task-2 Lab Report

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1. Objective

The objective of this task was to build a safe and isolated cybersecurity practice environment on a personal computer. This hands-on lab serves as the foundation for future tasks such as vulnerability assessment, web application testing, exploitation practice, and incident response.

The lab comprises the following components:

- 1 Attacker machine – Kali Linux (running inside VirtualBox)
 - 1 Vulnerable target application – OWASP Juice Shop (deployed via Docker on Kali Linux)
 - Proper network segmentation – NAT adapter for internet updates; Host-Only for isolated lab traffic
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2. Environment Overview

Component	Details
Host Operating System	Windows (running VirtualBox)
Virtualization Software	Oracle VirtualBox (latest)
Attacker VM	Kali Linux (kali@kali)
Target Application	OWASP Juice Shop v17 (Docker container)
Target URL	http://localhost:3000
Network Mode	NAT (internet) + Host-Only (lab traffic)
Tools Used	Burp Suite Community, Docker, Nmap (planned)

3. Step-by-Step Lab Build

Step 1 – Install VirtualBox on Host Machine

VirtualBox was downloaded from the official website and installed on the Windows host. During setup, a Host-Only network adapter was created and NAT was enabled to allow internet access from within the VMs.

Step 2 – Deploy Kali Linux (Attacker VM)

The pre-built Kali Linux VirtualBox image was imported into VirtualBox. The VM was configured as follows:

- CPUs: 2 cores
- RAM: 3–4 GB
- Adapter 1: NAT (internet updates)
- Adapter 2: Host-Only (isolated lab traffic)

Screenshot Evidence: Image 3 shows the Oracle VirtualBox Manager interface on the Windows host, confirming successful installation and readiness to create/import VMs.

Step 3 – Deploy OWASP Juice Shop via Docker

Docker was installed on the Kali Linux VM and the OWASP Juice Shop container was pulled and launched. The exact commands executed were:

```
sudo apt install -y docker.io
sudo systemctl enable --now docker
sudo docker pull bkimminich/juice-shop
sudo docker run -d -p 3000:3000 --name juice bkimminich/juice-shop
```

Screenshot Evidence: Image 4 (terminal screenshot) confirms the Docker pull of bkimminich/juice-shop executing on the Kali Linux VM, with all image layers completing the Pull Complete status. The kali-menu 2026.2.4 trigger confirms this is a fresh, up-to-date Kali installation.

Step 4 – Verify Target Access in Browser

After the Docker container started, the OWASP Juice Shop was accessed in the browser at <http://localhost:3000>. The application loaded successfully, displaying the product listing page with the cookie consent banner — confirming the target was live and ready for testing.

Screenshot Evidence: Image 5 shows the OWASP Juice Shop home page running at `http://localhost:3000` inside the Kali Linux VM's browser, with products (Apple Juice, Apple Pomace, etc.) displayed and the cookie banner visible.

Step 5 – Basic Validation with Burp Suite

Burp Suite Community Edition v2026.3.2 was launched on the Kali Linux VM. The browser proxy was configured to route traffic through Burp. The Logger tab was used to capture and inspect all HTTP traffic generated when browsing the Juice Shop application.

Key observations from the Logger:

- All requests were GET requests to localhost on port 3000
- Static assets loaded successfully (JavaScript chunks, CSS, images) with HTTP 200 status codes
- API endpoints were visible: `/api/Challenges/`, `/rest/products/search`, `/api/Quantities/`, `/rest/admin/application-configuration`
- HTTP 304 Not Modified responses were observed for cached admin configuration endpoints
- Response sizes ranged from 12 bytes to over 784 KB (main.js)
- Socket.IO polling was captured on `/socket.io/` with 3 query parameters — confirming real-time connectivity

Screenshot Evidence: Image 1 shows the Burp Suite Logger tab with 30 captured requests, all targeting localhost, confirming successful proxy intercept and traffic analysis.

4. Tools Used and Validation

Tool	Purpose	Status
Oracle VirtualBox	Host virtualization — running Kali Linux VM	Installed & Configured
Kali Linux (kali@kali)	Attacker VM — main penetration testing OS	Running
Docker	Container platform to run Juice Shop target	Active
OWASP Juice Shop	Intentionally vulnerable web application target	Live on port 3000
Burp Suite Community v2026.3.2	HTTP proxy for traffic interception and analysis	Capturing traffic
Nmap	Network scanning and host discovery (planned)	Available in Kali

5. Network Architecture

The lab is configured with two network adapters on the Kali VM to ensure proper isolation:

Adapter	Mode	Purpose
Adapter 1	NAT	Allows Kali VM to access the internet for package updates and tool downloads
Adapter 2	Host-Only	Creates an isolated internal network for lab traffic between the host and VMs

The OWASP Juice Shop runs inside Docker on the Kali VM and is accessible at `http://localhost:3000` from within the VM itself. Burp Suite intercepts traffic between the browser and the target on this loopback interface.

6. Observations from Initial Traffic Analysis

During the Burp Suite logging session, several interesting API endpoints and behaviors were noted that indicate areas for further testing:

Endpoint	Method	Status	Observation
/api/Challenges/	GET	200	Returns challenge list — indicates CTF-style challenge tracking
/rest/products/search	GET	200	Query parameter 'q=' present — potential SQL injection surface
/rest/admin/application-configuration	GET	304	Admin config endpoint accessible — potential info disclosure
/rest/admin/application-version	GET	304	Application version exposed — useful for vulnerability mapping
/api/Quantities/	GET	200	Inventory quantities endpoint — potential for manipulation
/socket.io/	GET	200	Real-time socket with EIO and transport params — live connection confirmed

7. Additional Tool Note – Postman API Test

A secondary test was performed using Postman to interact with an AI audit API (unrelated to Juice Shop). Image 2 shows a POST request with a JSON payload containing workflow, team satisfaction, pain points, highest cost, team size, and optimization goal fields. The server returned HTTP 500 Internal Server Error with the message 'Audit generation failed'.

Additionally, Postman flagged a code warning in the Post-request Script: the use of a Unicode en-dash (U+2013) where an ASCII hyphen (U+002D) was expected. This was noted for future debugging.

8. Deliverables Checklist

Deliverable	Status	Evidence
VirtualBox installed on host	Complete	Image 3 – VirtualBox Manager open on Windows
Kali Linux VM running	Complete	Image 4 – kali@kali terminal, Docker pull active
OWASP Juice Shop deployed via Docker	Complete	Image 4 – docker pull; Image 5 – browser at localhost:3000
Target accessible in browser	Complete	Image 5 – Juice Shop product page loaded
Burp Suite intercepting traffic	Complete	Image 1 – Burp Logger with 30 captured requests
IP configuration (NAT + Host-Only)	Complete	Configured in VirtualBox settings
Lab report (this document)	Complete	This PDF/DOCX report

9. Conclusion

The personal cybersecurity lab has been successfully built and validated. All five steps from the task document were completed:

- VirtualBox installed and configured on the Windows host machine
- Kali Linux VM imported and running with dual-adapter network (NAT + Host-Only)
- OWASP Juice Shop deployed via Docker and accessible at <http://localhost:3000>
- Browser traffic successfully proxied through Burp Suite Community Edition
- Initial API surface mapped through Burp Logger analysis

The lab environment is now fully operational and ready for the next tasks, including vulnerability scanning with Nmap, web application penetration testing on OWASP Juice Shop, and traffic analysis with Wireshark.

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